

Ruijie Xu

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EDUCATION

Georgetown University | 2024 – 2026 | Master of Science in Data Science & Analytics | GPA: 3.7 / 4.0

Soochow University | 2018 – 2023 | Bachelor of Architecture | GPA: 3.5 / 4.0

RESEARCH PROJECTS

Data Center Siting Analysis and Predictive Modeling (Ongoing)

- Developed two web scrapers to collect 4,428 data center locations across 51 states and extract county-level policy information, identifying 80+ candidate counties via frequency-based filtering
- Built an LLM-based policy classifier using OpenAI API to label county regulations, with human-in-the-loop validation for ambiguous cases
- Integrated 10+ official datasets into a unified county-level working table with 38 engineered features, keyed by FIPS codes
- Developing baseline regression models to evaluate predictive signals across engineered features

Translation Quality Estimation for Hindi-Chinese Pairs

- Refactored a 1,000+ line monolithic notebook into a modular, reproducible ML pipeline architecture
- Scaled training dataset from 100K to 800K sentence pairs and conducted cross-validation across 76 hyperparameter configurations spanning three model families, improving final test R^2 by +0.02
- Performed SHAP-based feature attribution analysis to identify key predictive signals and compared model behaviors across architectures

Distributed Reddit Data Processing & Analytics (Spark-Based)

- Processed 120M+ Reddit records (~120GB parquet) using PySpark to build scalable ETL and feature pipelines
- Engineered NLP features (TF-IDF, token statistics, length metrics) and conducted exploratory analysis on award patterns and high-engagement posts
- Trained and tuned logistic regression models on balanced samples; achieved AUC = 0.72 on the full imbalanced dataset

SnapStyle: CLIP-Based Cross-Modal Retrieval & Deployment System

- Integrated CLIP to encode detected clothing items into embeddings and constructed a FAISS vector database for efficient similarity search
- Built natural language query pipeline by embedding user inputs and performing approximate nearest neighbor (ANN) search against the clothing vector index
- Designed adjustable weight-based post-retrieval ranking mechanism and deployed the multi-service system via Docker and Hugging Face

LLM Validation Agent with Tiered Escalation (Ongoing)

- Prototyping multi-stage validation pipeline that escalates low-confidence outputs to higher-cost models for reliability control
- Integrating lightweight validation checkpoints within agent loops to detect inconsistencies before escalating to stronger LLM validators

EXPERIENCE

Teaching Assistant | Statistical Learning | Georgetown University | 2026

- Reviewed implementations of gradient descent, regularized regression, and classification models
- Analyzed common convergence and data preprocessing errors across student submissions
- Applied standardized grading rubrics to ensure consistency across assignments and exams

SKILLS

Programming: Python | SQL | R | Bash

Machine Learning: Supervised Models | Feature Engineering | Model Evaluation | Transformers

Generative AI & LLM Systems: RAG | Prompt Engineering

Distributed Computing: Apache Spark (PySpark)

Tools: Docker | Git | Jupyter

LANGUAGES

English (Proficient) | Mandarin (Native)